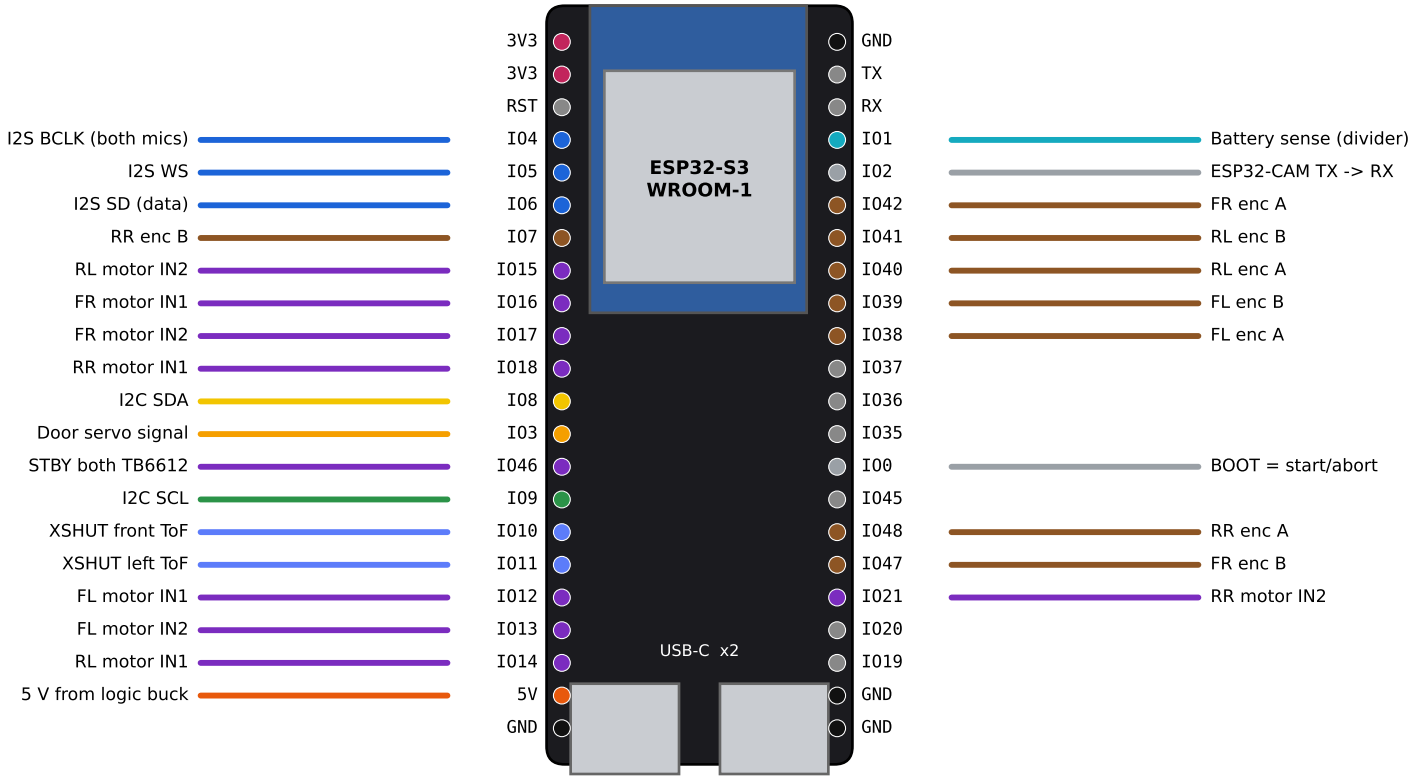
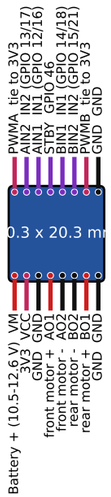
A.E.G.I.S. wiring guide 1/3: ESP32-S3-DevKitC-1 pin assignment (top view, antenna up)

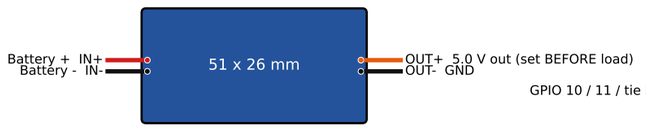


A.E.G.I.S. wiring guide 2/3: module faces, pins and where each wire lands (top views, to scale)

TB6612FNG (x2: A = left wheels, B = right)



XL4015 buck (x2: logic 5.0 V, servo 5.0 V)



GY-530 VL53L0X (x3: front, left, right)



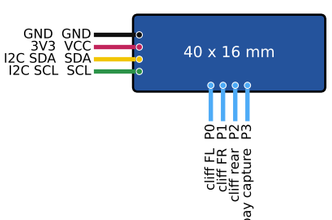
GY-521 MPU-6050



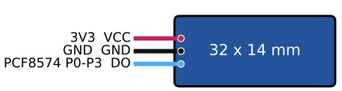
INMP441 mic (x2)



PCF8574T I/O expander



TCRT5000 / IR obstacle module (x4)



MG946R servo | N20 encoder motor (x4)



Battery 12 V
Ground
5 V logic
5 V servo
3.3 V
I2C SDA
I2C SCL
I2S audio
Motor PWM / STBY
Encoders
ToF XSHUT
IR sensors
UART / misc
Battery sense

A.E.G.I.S. wiring guide 3/3: wire list

From	To	Signal	Wire
Battery + (DC jack lead)	Both XL4015 IN+, both TB6612 VM	10.5-12.6 V	20 AWG
Battery -	Both XL4015 IN-, all GND	0 V	20 AWG
XL4015 logic OUT+	ESP32-S3 5V, ESP32-CAM 5V	5.0 V	22 AWG
XL4015 servo OUT+	MG946R red	5.0 V	22 AWG
ESP32-S3 3V3	sensors VCC, TB6612 VCC + PWMA/PWMB, encoders	3.3 V	26 AWG
ESP32 GPIO 8 / 9	VL53L0X x3, MPU-6050, PCF8574 (SDA/SCL), 2.2 k pull-ups	I2C 400 kHz	26 AWG
ESP32 GPIO 4 / 5 / 6	both INMP441 SCK / WS / SD	I2S 48 kHz	26 AWG, short
ESP32 GPIO 10 / 11	front / left VL53L0X XSHUT (right: tie to VIN)	address select	26 AWG
ESP32 GPIO 12-18, 21	TB6612 AIN1/AIN2/BIN1/BIN2	20 kHz PWM	26 AWG
ESP32 GPIO 46	STBY on both TB6612	enable	26 AWG
TB6612 AO1/AO2/BO1/BO2	motor red / white	motor power	24 AWG
Encoders yellow / green	GPIO 38/39, 40/41, 42/47, 48/7	quadrature	26 AWG
ESP32 GPIO 3	MG946R orange	50 Hz servo pulse	26 AWG
ESP32-CAM U0T	ESP32 GPIO 2	UART 115200	26 AWG
Battery + via 100k / 22k divider	ESP32 GPIO 1	battery voltage	26 AWG
IR modules DO	PCF8574 P0-P3	cliff / capture	26 AWG

Common ground star point at the battery negative. Set both XL4015 outputs to 5.0 V with a meter before connecting any board. Twist each motor's two leads together. Keep the I2S wires short (< 15 cm).
Header orders follow the reference boards; clone modules sometimes differ: check the silkscreen before soldering.